

Member ID: _____

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SQL Database Fundamentals

(345)

STATE 2025

Multiple Choice:

50 @ 2 points each

_____ (100 points)

Test Time: 60 minutes

Multiple Choice Questions

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. Which of the following backup terms does not fit with the others?
 - A. Hot backup
 - B. Cold backup
 - C. Online backup
 - D. Dynamic backup
2. A view built from two or more different tables can have which of the following properties?
 - A. Can be updated directly
 - B. Can be deleted without affecting underlying data
 - C. Will make database maintenance easier
 - D. Limits database overhead
3. What is the primary purpose of the SQL CASCADE option in a foreign key constraint?
 - A. Restricts changes in the parent table
 - B. Creates a backup of the database
 - C. Automatically generates unique values for the foreign key
 - D. Updates or deletes related records in child tables when changes occur in the parent table
4. Which SQL statement is used to add a new column to an existing table?
 - A. UPDATE COLUMN
 - B. MODIFY COLUMN
 - C. INSERT COLUMN
 - D. ADD COLUMN
5. What does CRUD stand for in the context of database operations?
 - A. Create, Recover, Update, Delete
 - B. Create, Read, Update, Debug
 - C. Create, Read, Update, Delete
 - D. Configure, Read, Update, Delete
6. What does ACID stand for in the context of database transactions?
 - A. All Command Integrity Declaration
 - B. Atomicity, Consistency, Isolation, Durability
 - C. Access Control and Identity Declaration
 - D. Automated Concurrency and Isolation Definition

7. What is normalization in database design?
 - A. Reducing redundancy and dependency by organizing fields and tables
 - B. Increasing redundancy for faster retrieval
 - C. Randomizing data for better security
 - D. Adding unnecessary fields for future use
8. Which of the following best describes a data warehouse?
 - A. A database optimized for online transaction processing (OLTP)
 - B. A database optimized for analysis and reporting of large volumes of data
 - C. A database used for temporary storage of data
 - D. A database used for real-time data processing
9. What does the term "OLAP" stand for in the context of databases?
 - A. Online Analysis Process
 - B. Online Analytical Processing
 - C. Offline Analytical Process
 - D. Offline Analytical Processing
10. What is a composite key in a database?
 - A. A key that consists of multiple attributes
 - B. A key that is generated automatically by the system
 - C. A key that is unique across all tables in the database
 - D. A key that is used for sorting records
11. Which of the following is an example of a NoSQL database?
 - A. PostgreSQL
 - B. Oracle Database
 - C. MongoDB
 - D. SQLite
12. What does the term "SQL injection" refer to?
 - A. A technique for optimizing SQL queries
 - B. A security vulnerability that allows attackers to execute malicious SQL code
 - C. A method for encrypting SQL data
 - D. A mechanism for creating temporary SQL tables
13. Which of the following is NOT a characteristic of NoSQL databases?
 - A. Structured schema
 - B. Scalability
 - C. High availability
 - D. Flexibility in data storage formats

14. Which of the following is NOT a type of database constraint?
- A. Primary key constraint
 - B. Foreign key constraint
 - C. Null constraint
 - D. Dynamic constraint
15. What does the term "database schema" represent?
- A. A physical storage location for database files
 - B. A predefined set of rules governing data integrity
 - C. The structure that defines the organization of data in a database
 - D. A mechanism for restricting access to certain database objects
16. What is the typical workload pattern in OLTP systems?
- A. Consistent and predictable
 - B. Highly variable with peaks and troughs
 - C. Intermittent with long periods of inactivity
 - D. Linearly increasing over time
17. What is the purpose of using table aliases in SQL joins?
- A. To limit the number of rows returned by the join operation
 - B. To enforce referential integrity constraints between joined tables
 - C. To avoid ambiguity when referencing columns from multiple tables
 - D. To improve query performance by optimizing the join algorithm
18. Which of the following operations is typically performed using DML?
- A. Creating tables
 - B. Modifying table structure
 - C. Inserting records into a table
 - D. Deleting tables
19. What does the acronym BCNF stand for in terms of database normalization?
- A. Binary Code Normal Form
 - B. Basic Complete Normal Form
 - C. Balanced Centralized Normal Form
 - D. Boyce-Codd Normal Form
20. What is the key constraint for the fourth normalization form?
- A. It requires every determinant to be a candidate key
 - B. It ensures no multiple independent multi-valued facts about an entity in a record
 - C. It eliminates all non-temporal redundancy
 - D. It eliminates transitive dependency

21. What is a transitive functional dependency?
- A. Changing a non-key column might cause another non-key column to change
 - B. When there exists more than one candidate key
 - C. It ensures rows in one table have corresponding rows in another
 - D. A primary key is functional dependent on a subset of candidate key relationships
22. In a multi-user database system, what does the term "concurrency control" refer to?
- A. The ability to process transactions independently of each other
 - B. The ability to ensure that transactions are processed as a single unit of work
 - C. The ability to ensure that changes made by one transaction do not interfere with changes made by other transactions
 - D. The ability to maintain the consistency of the database before and after a transaction
23. Which of the following is an example of a transitive dependency?
- A. A customer's name and address
 - B. A product's price and quantity
 - C. A student's ID, course, and instructor
 - D. A department's name and location
24. What is a "dirty read" in the context of database transactions?
- A. A read operation that retrieves data from the database without applying any locks
 - B. A read operation that retrieves uncommitted data changes made by another transaction
 - C. A read operation that retrieves inconsistent data due to a database corruption
 - D. A read operation that retrieves data from a table without any specified criteria
25. Which of the following is NOT a level of isolation defined by the SQL standard?
- A. READ UNCOMMITTED
 - B. READ COMMITTED
 - C. REPEATABLE READ
 - D. WRITE PENDING
26. In which isolation level are dirty reads allowed in most database systems?
- A. READ UNCOMMITTED
 - B. READ COMMITTED
 - C. REPEATABLE READ
 - D. WRITE PENDING

27. Given user A reads a record and user B inserts or deleted rows to A's record. If user A reads the same rows again, a new row is found. This is known as what?
- A. A Dirty Read
 - B. A non-Repeatable Read
 - C. A Lost Read
 - D. A Phantom Read
28. Which database locking mechanism allows multiple transactions to read a data item simultaneously but prevents any transaction from writing to it?
- A. Shared lock
 - B. Exclusive lock
 - C. Intent lock
 - D. Update lock
29. What is an advantage of using optimistic locking over pessimistic locking?
- A. Reduced risk of deadlock
 - B. Higher level of data consistency
 - C. Improved concurrency and performance
 - D. Enhanced data durability
30. In which scenario would you choose to use a stored procedure instead of an ad-hoc SQL query executed from an application?
- A. When the SQL query needs to be executed only once
 - B. When the SQL query needs to be executed multiple times with different parameters
 - C. When the SQL query involves complex business logic or multiple steps
 - D. When the SQL query requires dynamic table creation or alteration
31. How can stored procedures improve database performance compared to executing ad-hoc SQL queries from an application?
- A. Stored procedures cache execution plans, reducing query compilation overhead
 - B. Stored procedures enforce stronger data consistency rules
 - C. Stored procedures automatically optimize SQL statements for better performance
 - D. Stored procedures eliminate the need for indexes on database tables
32. What is the difference between a stored procedure and a user-defined function (UDF) in a database?
- A. Stored procedures can only return scalar values, whereas UDFs can return result sets
 - B. Stored procedures can execute DML statements, whereas UDFs cannot
 - C. Stored procedures can only be invoked by the database administrator, whereas UDFs can be invoked by any user
 - D. Stored procedures are precompiled and stored in the database, whereas UDFs are compiled at runtime

33. Which of the following statements accurately describes dynamic SQL within a stored procedure?
- A. Dynamic SQL allows stored procedures to be executed without compilation
 - B. Dynamic SQL is a feature that enables the creation of temporary tables within a stored procedure
 - C. Dynamic SQL allows stored procedures to construct and execute SQL statements at runtime
 - D. Dynamic SQL provides built-in error handling mechanisms for stored procedures
34. A key that is invented or made up for the sole purpose of being used as the primary key is known as what type of key?
- A. Natural Key
 - B. Composite Key
 - C. Surrogate Key
 - D. Unique Key
35. Which of the following operations is NOT typically performed using DDL?
- A. Deleting tables
 - B. Inserting records into a table
 - C. Creating tables
 - D. Modifying table structure
36. In a database ER diagram, what does a solid line connecting two entity sets represent?
- A. A one-to-many relationship
 - B. A many-to-many relationship
 - C. A one-to-one relationship
 - D. An associative relationship
37. What does DBaaS stand for?
- A. Distributed Business as a Service
 - B. Database as a Service
 - C. Digital Backup as a Service
 - D. Data Backup and Storage
38. Which of the following is a typical use case for DBaaS?
- A. High-performance computing
 - B. Small-scale personal websites
 - C. On-premises legacy systems
 - D. Enterprise-level data warehousing

39. What is a key benefit of using a managed DBaaS solution?
- A. Increased need for in-house database expertise
 - B. Enhanced control over infrastructure
 - C. Reduced operational overhead
 - D. Longer deployment times
40. What is polyglot persistence?
- A. Storing data in multiple languages for internationalization purposes
 - B. Using multiple database management systems within a single application
 - C. Persisting data in a single database system to maintain consistency
 - D. Utilizing multiple programming languages for database interaction
41. Which challenge is commonly associated with maintaining polyglot persistence?
- A. Increased development complexity
 - B. Reduced data consistency
 - C. Limited scalability options
 - D. Lower performance across all databases
42. Which query optimization technique involves restructuring SQL queries to produce the same result more efficiently?
- A. Query parallelization
 - B. Query caching
 - C. Query rewriting
 - D. Query indexing
43. How does query parallelization contribute to database optimization?
- A. By executing multiple queries in sequence to reduce resource consumption
 - B. By executing multiple queries concurrently to improve performance
 - C. By optimizing query execution plans for complex queries
 - D. By limiting the number of concurrent database connections
44. What is schema-on-read in the context of data lakes?
- A. Data is stored in a structured format before being read by applications
 - B. Data is stored without a predefined schema and is structured at the time of analysis
 - C. Data is transformed into a standard schema before being stored in the data lake
 - D. Data is encrypted and secured before being read by applications
45. What is the purpose of database auditing?
- A. To encrypt data stored in the database
 - B. To monitor and record database activity for security and compliance purposes
 - C. To restrict access to certain database objects based on user roles
 - D. To authenticate users before allowing them to access the database

46. Which property of BASE emphasizes that a system continues to operate even in the presence of partial failures?
- A. Basic Availability
 - B. Soft state
 - C. Eventually consistent
 - D. Scalability
47. Which aspect of BASE ensures that system performance remains stable as load increases?
- A. Basic Availability
 - B. Soft state
 - C. Eventually consistent
 - D. Scalability
48. What does data lineage tracing involve?
- A. Encrypting data during transmission
 - B. Tracing the movement and transformation of data across systems and processes
 - C. Auditing user access to sensitive data
 - D. Validating data consistency within a database
49. What is one of the biggest challenges associated with data migration?
- A. Ensuring data consistency across systems
 - B. Implementing encryption for data in transit
 - C. Managing access controls within the database
 - D. Optimizing database performance after migration
50. What is a weak entity in an ER model?
- A. An entity that does not have a primary key
 - B. An entity that relies on another entity for identification
 - C. An entity with a high degree of cardinality
 - D. An entity that contains only derived attributes